

Cross-sectional study.

- E: Exposure (E.g. smoking, drinking...) $\begin{matrix} \text{E} \\ \text{E} \end{matrix}$
- D: Disease. $\begin{matrix} \text{D} \\ \text{D} \end{matrix}$

Data: (contingency table)

	D	$\bar{D}$	
E	$N_{11} = n_{11}$	$N_{12} = n_{12}$	n_{1+}
$\bar{E}$	$N_{21} = n_{21}$	$N_{22} = n_{22}$	n_{2+}
	n_{+1}	n_{+2}	n_{++} (Fixed)

R.V.: $(N_{11}, N_{12}, N_{21}, N_{22}) \sim \text{Multinomial}(n_{++}, p_1, p_2, p_3, p_4)$

$$\begin{aligned} \text{Var}(N_{ij}) &= n_{++} p_i (1 - p_i) \\ \text{Cov}(N_{ij}, N_{i'j'}) &= -n_{++} p_i p_{i'} \\ E(N_{ij}) &= n_{++} \cdot p_i \\ \sum p_i &= 1 \end{aligned}$$

$$\begin{aligned} p_1 &= P(D, E) & \hat{p}_1 &= \frac{n_{11}}{n_{++}} \\ p_2 &= P(\bar{D}, E) & \hat{p}_2 &= \frac{n_{12}}{n_{++}} \\ p_3 &= P(D, \bar{E}) & \hat{p}_3 &= \frac{n_{21}}{n_{++}} \\ p_4 &= P(\bar{D}, \bar{E}) & \hat{p}_4 &= \frac{n_{22}}{n_{++}} \end{aligned}$$

Prevalence of disease

$$\begin{aligned} \hat{P}(D) &= \hat{P}(D, E) + \hat{P}(D, \bar{E}) \\ &= \frac{n_{11} + n_{21}}{n_{++}} = \frac{n_{+1}}{n_{++}} \end{aligned}$$

Prob of exposure

$$\begin{aligned} \hat{P}(E) &= \hat{P}(D, E) + \hat{P}(\bar{D}, E) \\ &= \frac{n_{11} + n_{12}}{n_{++}} = \frac{n_{1+}}{n_{++}} \end{aligned}$$

$$\Rightarrow \text{Relative risk} = \psi = \frac{P(D|E)}{P(D|\bar{E})}$$

$$\frac{\hat{P}(D|E)}{\hat{P}(D|\bar{E})} = \frac{\frac{n_{11}}{n_{1+}}}{\frac{n_{21}}{n_{2+}}} = \hat{\psi} \begin{cases} > 1: \text{positive association} \\ = 1: \text{no association} \\ < 1: \text{negative association} \end{cases}$$

$$\Rightarrow \text{Odds ratio} = \theta = \frac{P(D|E)/1-P(D|E)}{P(D|\bar{E})/1-P(D|\bar{E})}$$

CI: $(1-\alpha)$

$$\textcircled{1} \exp \left\{ \log(\hat{\theta}) \pm Z_{1-\frac{\alpha}{2}} \sqrt{\text{Var}\{\log(\hat{\theta})\}} \right\}$$

$$\hat{\text{Var}}\{\hat{\theta}\} = \text{Var}\{\exp\{\log(\hat{\theta})\}\}$$

$$= \frac{d \exp\{\log(\hat{\theta})\}}{d \log(\hat{\theta})} \cdot \text{Var}\{\log(\hat{\theta})\}$$

$$= \{\log(\hat{\theta})\}^2 \cdot \hat{\text{Var}}\{\log(\hat{\theta})\}$$

$$= \hat{\theta}^2 \cdot \left\{ \frac{1}{n_{11}} + \frac{1}{n_{22}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} \right\}$$

$$\textcircled{2} \hat{\theta} \pm Z_{1-\frac{\alpha}{2}} \sqrt{\hat{\text{Var}}(\hat{\theta})}$$

Rare event. $P(D|E) \approx 0 \Rightarrow n_{12} \approx n_{1+}$

$P(D|\bar{E}) \approx 0 \Rightarrow n_{22} \approx n_{2+}$

$\Rightarrow \theta \approx \psi$ *Better interpretation*

$$\frac{\hat{P}(D|E)/1-\hat{P}(D|E)}{\hat{P}(D|\bar{E})/1-\hat{P}(D|\bar{E})} = \frac{n_{11} \cdot n_{22}}{n_{12} \cdot n_{21}} = \hat{\theta} \begin{cases} > 1: \text{positive association} \\ = 1: \text{no association} \\ < 1: \text{negative association} \end{cases}$$

$$\log OR: \log \hat{\theta} = \log(n_{11}) + \log(n_{22}) - \log(n_{12}) - \log(n_{21})$$

$$\hat{\text{Var}}\{\log(\hat{\theta})\} = \text{Var}\{\log(n_{11})\} + \text{Var}\{\log(n_{22})\} + \text{Var}\{-\log(n_{12})\} + \text{Var}\{-\log(n_{21})\}$$

$$= \left\{ \frac{d \log(n_{11})}{d n_{11}} \right\}^2 \hat{\text{Var}}(n_{11}) + \dots$$

$$= \frac{1}{n_{11}^2} \cdot n_{11} + \frac{1}{n_{22}^2} \cdot n_{22} + \frac{1}{n_{12}^2} \cdot n_{12} + \frac{1}{n_{21}^2} \cdot n_{21}$$

$$= \frac{1}{n_{11}} + \frac{1}{n_{22}} + \frac{1}{n_{12}} + \frac{1}{n_{21}}$$

About $\text{Var}(n_{11})$: Assume. $n_{11} \sim \text{Poisson}(\lambda)$

$$E\{n_{11}\} = \text{Var}\{n_{11}\} \approx n_{11}$$

Assume. Binomial Distribution by fixing n_{1+} & n_{2+}

(Afterwards)

$$g(x_n) - g(\theta) = g'(\theta) \{x_n - \theta\}$$

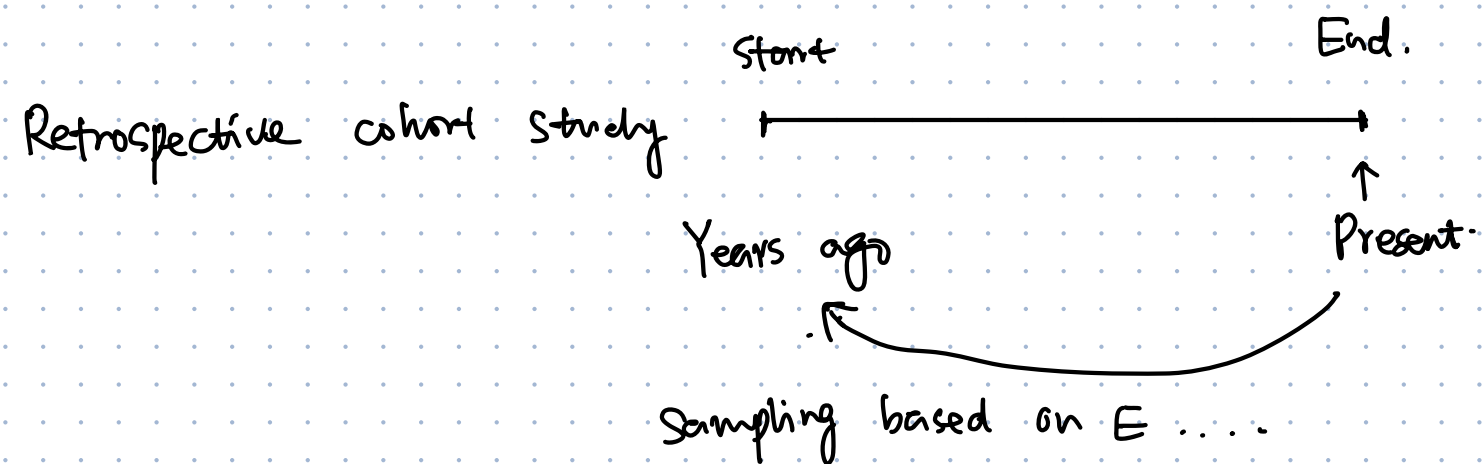
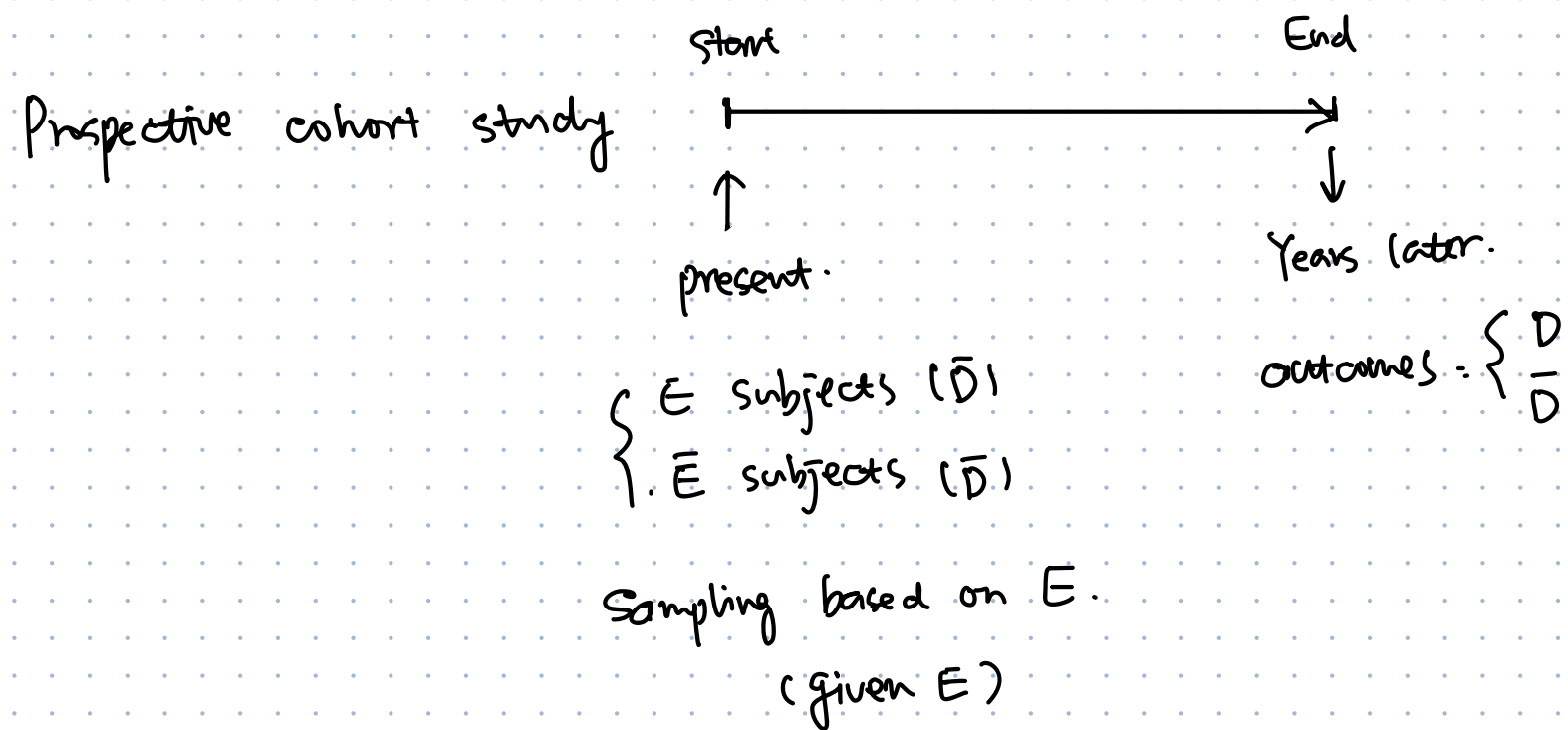
Delta method: (Taylor expansion)

$$g(x_n) = g(\theta) + g'(\theta) \{x_n - \theta\} + \dots$$

$$\sqrt{n} \{g(x_n) - g(\theta)\} = \sqrt{n} \{g'(\theta) [x_n - \theta]\}$$

$$\xrightarrow{D} N(0, \{g'(\theta)\}^2 \cdot \sigma^2) \quad \xrightarrow{D} N(0, \sigma^2)$$

Longitudinal Study. $\begin{cases} \text{Prospective} \\ \text{Retrospective} \end{cases}$



Data	D	$\bar{D}$	
E	$N_{11} = n_{11}$	n_{12}	$n_{1+} \leftarrow$
$\bar{E}$	$N_{21} = n_{21}$	n_{22}	$n_{2+} \leftarrow$

Fixed sample size.

↑
P.V. (N_{11}, N_{21})

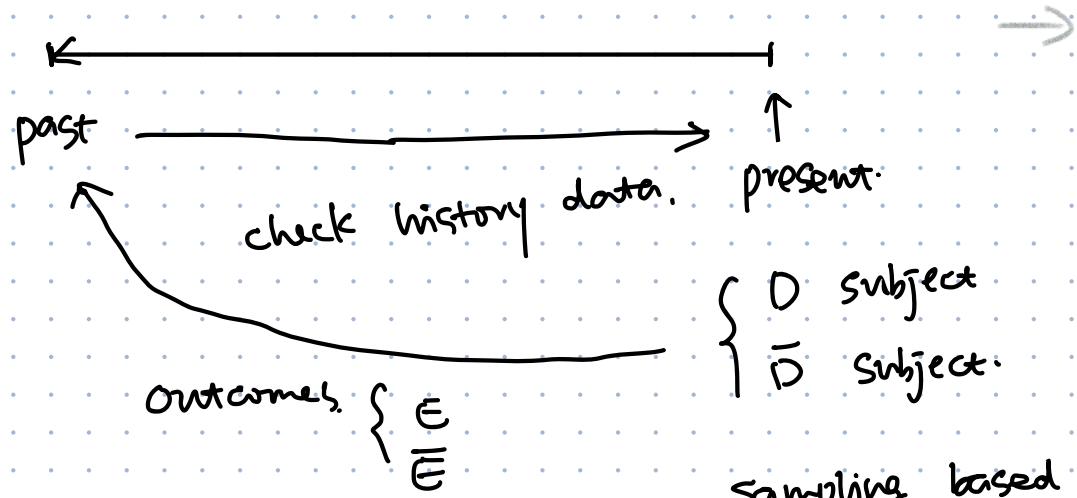
$$\begin{cases} N_{11} \sim \text{Bin}(n_{1+}, p_1) & p_1 = P(D|E) & \hat{p}_1 = \frac{n_{11}}{n_{1+}} \\ N_{21} \sim \text{Bin}(n_{2+}, p_2) & p_2 = P(D|\bar{E}) & \hat{p}_2 = \frac{n_{21}}{n_{2+}} \end{cases}$$

$$\hat{\psi} = \frac{n_{11} n_{2+}}{n_{1+} n_{21}}$$

$$\hat{\theta} = \frac{n_{11} n_{22}}{n_{12} n_{21}}$$

$$= \frac{P(D|E)/P(\bar{D}|E)}{P(D|\bar{E})/P(\bar{D}|\bar{E})}$$

Retrospective case-control study



Data:

	D	$\bar{D}$
E	$N_{11}=n_{11}$	$N_{12}=n_{12}$
$\bar{E}$	n_{21}	n_{22}
	n_{+1}	n_{+2}

R.V. (N_{11}, N_{12})

Fixed.

Odds Ratio

$$\begin{aligned} \theta &= \frac{P(D|E)/1-P(D|E)}{P(D|\bar{E})/1-P(D|\bar{E})} \\ &= \frac{P(D|E)}{P(\bar{D}|E)} \cdot \frac{P(\bar{D}|\bar{E})}{P(D|\bar{E})} = A \cdot B. \end{aligned}$$

$$\begin{aligned} \frac{P(D|E)}{P(D|\bar{E})} &= \frac{P(E|D) \cdot P(D)}{P(E)} \cdot \frac{P(\bar{E})}{P(\bar{E}|D)P(D)} = A \\ \frac{P(\bar{D}|\bar{E})}{P(D|\bar{E})} &= \frac{P(\bar{E}|\bar{D}) \cdot P(\bar{D})}{P(\bar{E})} \cdot \frac{P(E)}{P(E|\bar{D})P(\bar{D})} = B \end{aligned}$$

$$AB = \frac{P(E|D) \cdot P(\bar{E}|\bar{D})}{P(\bar{E}|D) \cdot P(E|\bar{D})} \quad \hat{\theta} = \frac{n_{11}n_{22}}{n_{12}n_{21}}$$

sampling based on D.
(given D)

$$\begin{cases} N_{11} \sim \text{Bin}(n_{+1}, p_1) & p_1 = P(E|D) & \hat{p}_1 = \frac{n_{11}}{n_{+1}} \\ N_{12} \sim \text{Bin}(n_{+2}, p_2) & p_2 = P(E|\bar{D}) & \hat{p}_2 = \frac{n_{12}}{n_{22}} \end{cases}$$

Relative risk $\psi = \frac{P(D|E)}{P(D|\bar{E})} \quad ??$

$$\begin{aligned} \hat{P}(D|E) &= \frac{n_{11}}{n_{1+}} \quad \text{NOT MAKE SENSE} \\ \hat{P}(D|\bar{E}) &= \frac{n_{21}}{n_{2+}} \end{aligned}$$

Consider : $n_{+2} = 50$; $n_{+1} \rightarrow \infty \Rightarrow n_{11} \uparrow \Rightarrow \hat{P}(D|E) = \frac{n_{11}}{n_{1+}} = \frac{n_{11}}{n_{11}+n_{12}} \rightarrow 1$
 $(p_1 > 0)$
 $n_{21} \uparrow \Rightarrow \hat{P}(D|\bar{E}) = \frac{n_{21}}{n_{2+}} = \frac{n_{21}}{n_{21}+n_{22}} \rightarrow 1$

Same event $\hat{\theta} \approx \hat{\psi}$